

MDP to do weekly assignment

Let the space space be 'S', action space be 'A', reward as 'R' and transition probability is 'T'.

S={ S0- Wake up , S1- Assignment completed, S2- Postponed}

A={ A1- Do assignment , A2- Sleep }

Let A1 has 70% chances to transitioning to S1 and 30% chances to S2. A2 has 90% chances for S2 and 10% for S1.

For taking A1 in initial state S0 ,R=+100; while for taking A2 ,R=10.

Discount factor (γ)=0.9

$$V^*(s) = \max_{a \in Ap(s)} \sum_{s' \in S} \mathcal{Pr}(s'|s, a) [\mathcal{R}(s, a, s') + \gamma V^*(s')]$$

Let V_0 for all S =0, value function for S1 and S2 remains 0 throughout as they are final states.

For S0:-

Action = A1 ;

$$V_1(S_0) = 0.7(100 + 0.9(0)) + 0.3(10 + 0.9(0)) = 73$$

Action = A2 ;

$$V1(S0) = 0.1(100 + 0.9(0)) + 0.9(10 + 0.9(0))=19$$

$$\Rightarrow V1(S0)=73$$

Action = A1 ;

$$V2(S0)= 0.7(100 + 0.9(73)) + 0.3(10 + 0.9(73))=138.7$$

Action = A2 ;

$$V2(S0) = 0.1(100 + 0.9(73)) + 0.9(10 + 0.9(73))=84.7$$

$$\Rightarrow V2(S0)=138.7$$

$\Rightarrow$ The ideal policy map is $V2(S0)=138.7$,

$$\Rightarrow V2(S1)=V2(S2)=0$$